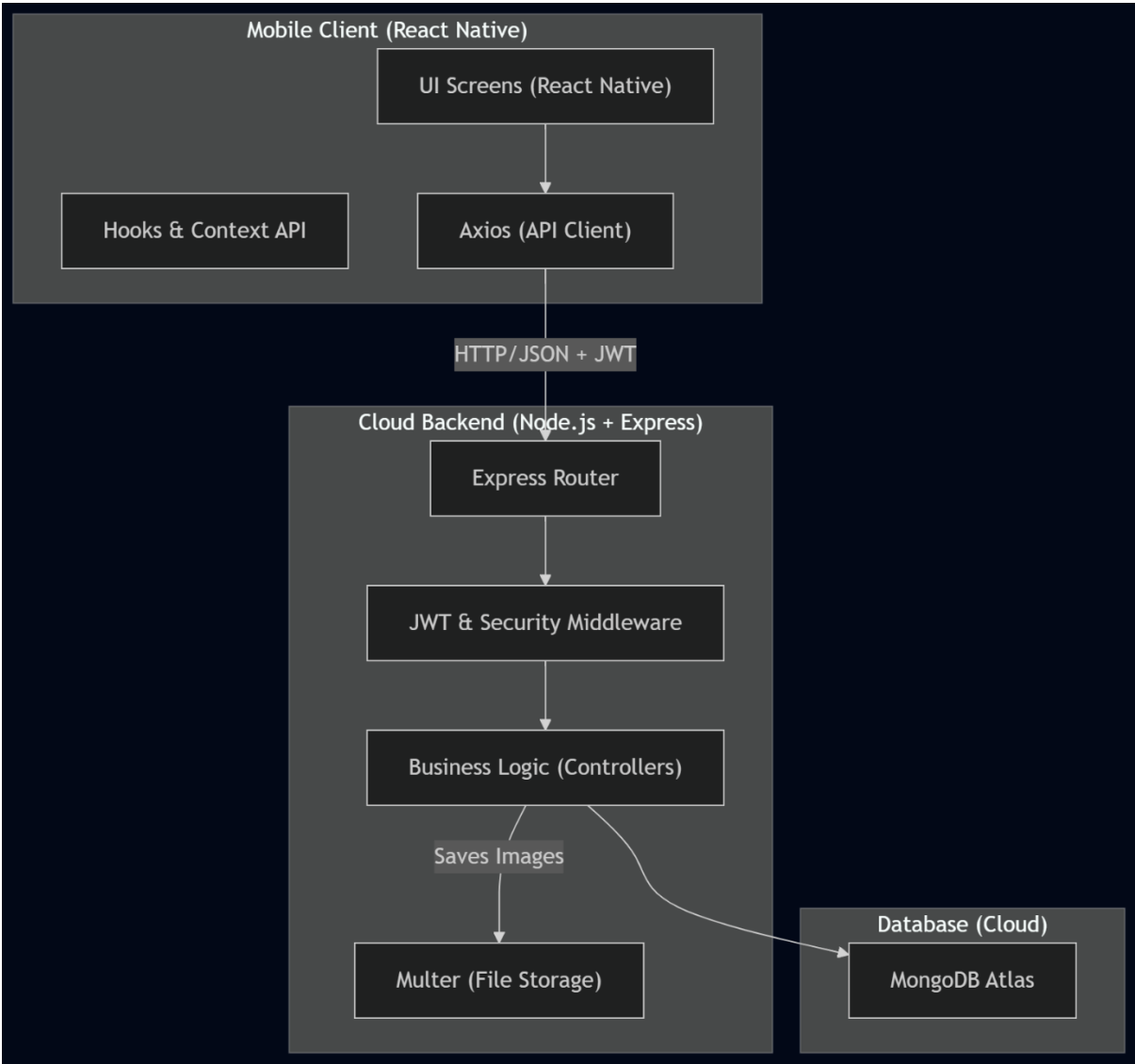


System Architecture

The Hostel Management System follows a classic **Full Stack MERN-style architecture** (replacing Web with React Native) to ensure scalability, security, and real-time data handling.

1. High-Level Architecture Diagram



2. Component Breakdown

A. Frontend (Mobile)

- **React Native / Expo:** The core framework for building the cross-platform UI.
- **Expo Router / Stack Navigation:** Manages the flow between Login, Dashboards, and Management screens.
- **Context API:** Manages global Authentication state (User roles, Tokens).
- **Axios:** Handles asynchronous communication with the hosted REST API.

B. Backend (REST API)

- **Node.js & Express:** Provides a robust server environment for handling concurrent requests.
- **JWT (JSON Web Tokens):** Securely handles user sessions and protects sensitive routes from unauthorized access.
- **Bcrypt:** Used for hashing user passwords before storage in the database.
- **Multer:** Handles file streaming for room images and payment proof uploads.

C. Database

- **MongoDB Atlas:** A cloud-hosted NoSQL database used for flexible data storage (JSON-like documents).
- **Mongoose:** An ODM (Object Data Modeling) library that enforces a schema at the application level.